

Vehicle Index

	Trackwidth	Wheelbase	Roof Height	Suspension Travel	Max Enging Torque	Driveline Gearing	Tire Diameter	Energy Capacity	Frame Rail Wall Thickness	Body Panel Thickness
1	196 in	80.2 in	79.3 in	22.9 in	728 lb-ft	5.87:1	46 in	1620 kWh	0.948 in	0.98 in
	Very Wide	Very Short	Tall Roof	Very Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Moderately Armored
2	110 in	80.1 in	65.5 in	20.8 in	421 lb-ft	2.15:1	49 in	1660 kWh	0.939 in	1.48 in
	Very Narrow	Very Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Heavily Armored
3	119 in	80.2 in	70.2 in	6.8 in	543 lb-ft	4.67:1	48 in	1560 kWh	0.285 in	0.18 in
	Very Narrow	Very Short	Short Roof	Very Short Travel	Moderate Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
4	181 in	81.2 in	69.2 in	9.7 in	655 lb-ft	2.37:1	32 in	1070 kWh	0.877 in	1.99 in
	Very Wide	Very Short	Short Roof	Short Travel	High Output	Very Long Gearing	Very Small Tire	Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
5	192 in	85.6 in	65.3 in	21.1 in	684 lb-ft	2.61:1	45 in	1240 kWh	0.783 in	1.48 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very High Output	Very Long Gearing	Large Tire	Moderate Energy Capacity	Heavy-Duty Frame	Heavily Armored
6	178 in	81.1 in	79.1 in	6.8 in	409 lb-ft	2.53:1	47 in	1530 kWh	0.251 in	0.47 in
	Wide	Very Short	Tall Roof	Very Short Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Light-Duty Frame	Very Lightly Armored
7	199 in	84.2 in	68.2 in	22.7 in	419 lb-ft	2.4:1	39 in	1520 kWh	0.331 in	1.74 in
	Very Wide	Short	Very Short Roof	Very Long Travel	Very Low Output	Very Long Gearing	Moderate Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
8	176 in	98.8 in	69.2 in	10.8 in	443 lb-ft	5.08:1	31 in	960 kWh	0.994 in	1.67 in
	Wide	Very Long	Short Roof	Short Travel	Very Low Output	Short Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Very Heavily Armored
9	195 in	96.4 in	66.1 in	9.3 in	630 lb-ft	5.12:1	50 in	1610 kWh	0.741 in	0.17 in
	Very Wide	Very Long	Very Short Roof	Very Short Travel	High Output	Short Gearing	Very Large Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
10	134 in	83.1 in	80 in	18.1 in	706 lb-ft	5.95:1	49 in	1010 kWh	0.512 in	1.75 in
	Narrow	Very Short	Tall Roof	Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very Low Energy Capacity	Light-Duty Frame	Very Heavily Armored
11	107 in	87.6 in	78 in	8.1 in	750 lb-ft	2.93:1	30 in	880 kWh	0.335 in	1.97 in
	Very Narrow	Short	Tall Roof	Very Short Travel	Very High Output	Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
12	114 in	99.7 in	82 in	18.4 in	715 lb-ft	2.25:1	30 in	980 kWh	0.967 in	0.74 in
	Very Narrow	Very Long	Very Tall Roof	Long Travel	Very High Output	Very Long Gearing	Very Small Tire	Very Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
13	107 in	98.2 in	66.6 in	23.1 in	512 lb-ft	5.32:1	30 in	1530 kWh	0.284 in	0.84 in
	Very Narrow	Very Long	Very Short Roof	Very Long Travel	Low Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
14	110 in	92 in	67.4 in	20.6 in	517 lb-ft	2.84:1	33 in	1210 kWh	0.432 in	1.82 in
	Very Narrow	Long	Very Short Roof	Very Long Travel	Low Output	Long Gearing	Very Small Tire	Moderate Energy Capacity	Light-Duty Frame	Very Heavily Armored

Vehicle Index

15	200 in	95.1 in	81.7 in	20.6 in	665 lb-ft	5.92:1	31 in	1460 kWh	0.313 in	1.67 in
	Very Wide	Long	Very Tall Roof	Very Long Travel	High Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
16	124 in	99.8 in	82.1 in	18.8 in	532 lb-ft	3.62:1	31 in	1570 kWh	0.782 in	0.33 in
	Narrow	Very Long	Very Tall Roof	Long Travel	Low Output	Moderately Gearing	Very Small Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
17	172 in	94 in	83.4 in	9.6 in	735 lb-ft	5.94:1	43 in	1500 kWh	0.313 in	0.78 in
	Wide	Long	Very Tall Roof	Very Short Travel	Very High Output	Very Short Gearing	Large Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
18	192.00527	98 in	82.4 in	6.5 in	442 lb-ft	3.23:1	35 in	1120 kWh	0.308 in	1.7 in
	Very Wide	Very Long	Very Tall Roof	Very Short Travel	Very Low Output	Long Gearing	Small Tire	Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
19	108 in	84.8 in	66.3 in	9.6 in	456 lb-ft	5.74:1	50 in	1080 kWh	0.966 in	0.84 in
	Very Narrow	Short	Very Short Roof	Very Short Travel	Very Low Output	Very Short Gearing	Very Large Tire	Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
20	102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
	Very Narrow	Long	Moderate Height Roof	Very Short Travel	Very High Output	Short Gearing	Large Tire	Very Low Energy Capacity	Heavy-Duty Frame	Very Lightly Armored

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

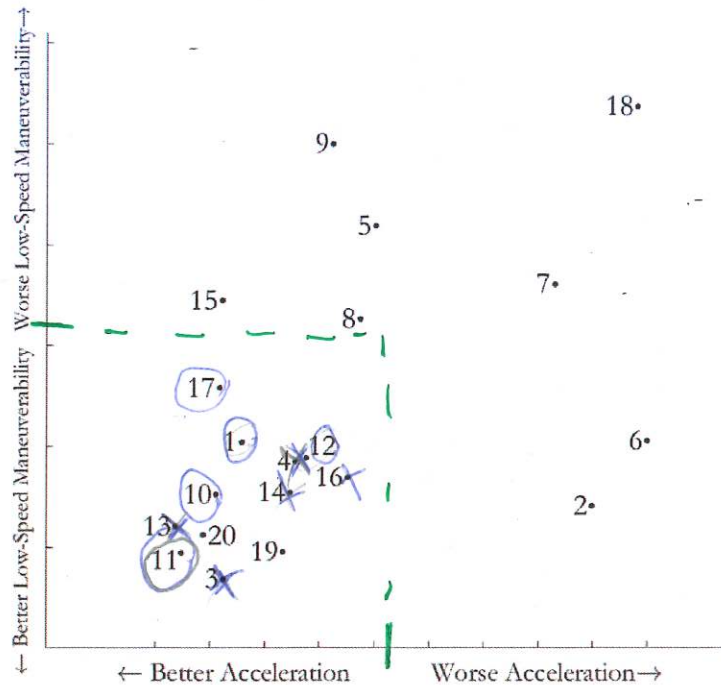
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Using Performance Space Plots

A set of 20 design configurations has been generated using the performance model. These configurations are plotted in performance space. All performance plots are framed so that utility increases the closer a point is to the origin. The axes are labeled with arrows indicating better or worse performance.



Each design is assigned an ID number. The design variables for each design are contained in the Vehicle Index (detached from the rest of the packet). It contains the design variable values and a verbal description of each vehicle.

	Trackwidth	Wheelbase	Roof Height	Suspension Travel	Max Enging Torque	Driveline Gearing	Tire Diameter	Energy Capacity	Frame Rail Wall Thickness	Body Panel Thickness
Vehicle 1	196	80.2	79.3	22.9	728	5.87	46	1619	0.948	0.98
	Very Wide	Very Short	Tall	Very Long Travel	Very High Output	Very Short Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Moderately Armored
Vehicle 2	110	80.1	65.5	20.8	421	2.15	49	1661	0.939	1.48
	Very Narrow	Very Short	Very Short	Very Long Travel	Very Low Output	Very Long Gearing	Very Large Tire	Very High Energy Capacity	Very Heavy-Duty Frame	Heavily Armored

Three rounds of design activity will follow. In some rounds, you will act according to your working group; in others, you will act as a single, cross-disciplinary team. In some rounds, you will have many performance space plots, in others, you will have performance space plots that only display performance objectives relevant to your working group.

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Round 1:

For this design exercise, you will be acting as a member of your assigned working group – you will have differing objectives and be provided performance space plots that DO NOT have any cross-disciplinary objective pairings.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

- A. Acceleration
- B. Towing Capacity - Torque

The Chassis and Cabin working group wishes to maximize the following objectives:

- C. Low Speed Maneuverability
- D. Passenger Cabin Space

Vehicle 17

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

110	92	67.4	20.6	517	2.84:1	33	1210	0.932	1.92
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*
*
*
*
*
(off-road
(top speed
(maintainability
(off-road, top speed
(protection

Do you think you could have found a better solution given more time?

Maybe given more time to analyze

What, in this round, stopped you from finding a better solution?

Time constraints

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Round 3:

For this design exercise, you will NOT be acting as a member of your assigned working group. You will act as a team that shares the same objectives.

INSTRUCTIONS:

Working as a cross-disciplinary team, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. Select a single vehicle from the 20 designs presented and write the selected design's ID in the space provided in the supplemental packet.

Your team wishes to maximize the following objectives:

- I. Top Speed -
- J. Operational Range -
- K. Occupant Protection -
- L. Rollover Stability -

7, 14, 7, 12, 4

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

110 in	92 in	67.7 in	23.6 in	517 lb ft	2.89 in	33 in	1200 lb	0.432 in	1.82 in
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- armor
- gear
- satisfies all of the constraints

Do you think you could have found a better solution given more time?

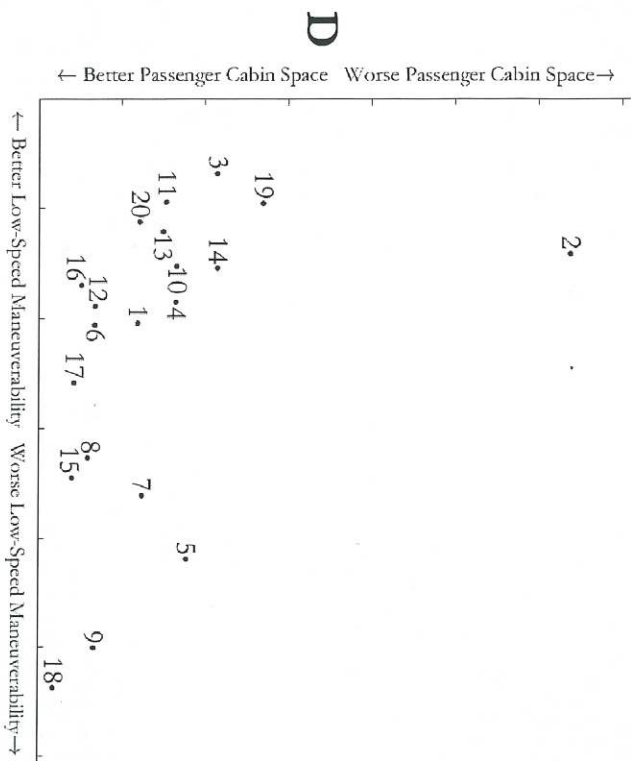
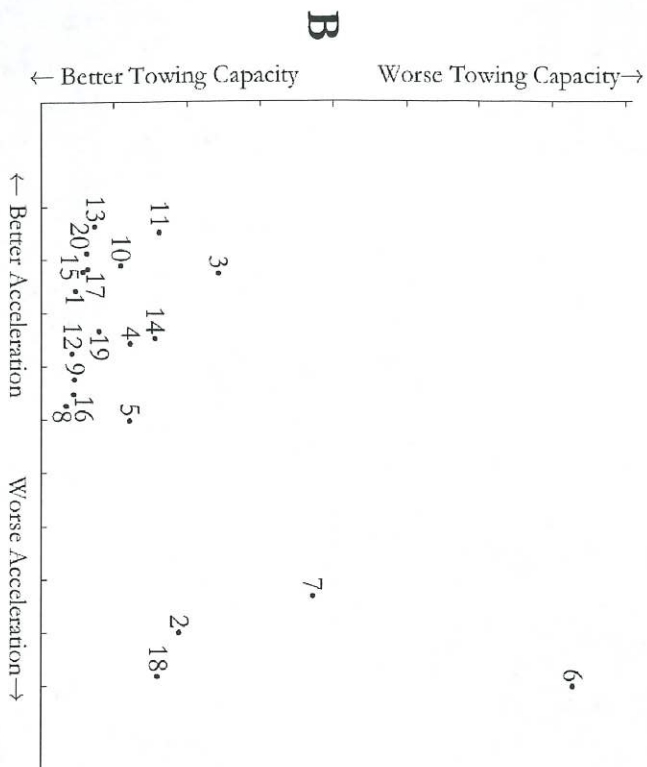
Possibly, if given more time to analyze the gages and spend more time discussing.

What, in this round, stopped you from finding a better solution?

- time
- limited information available
- limited data on requirements

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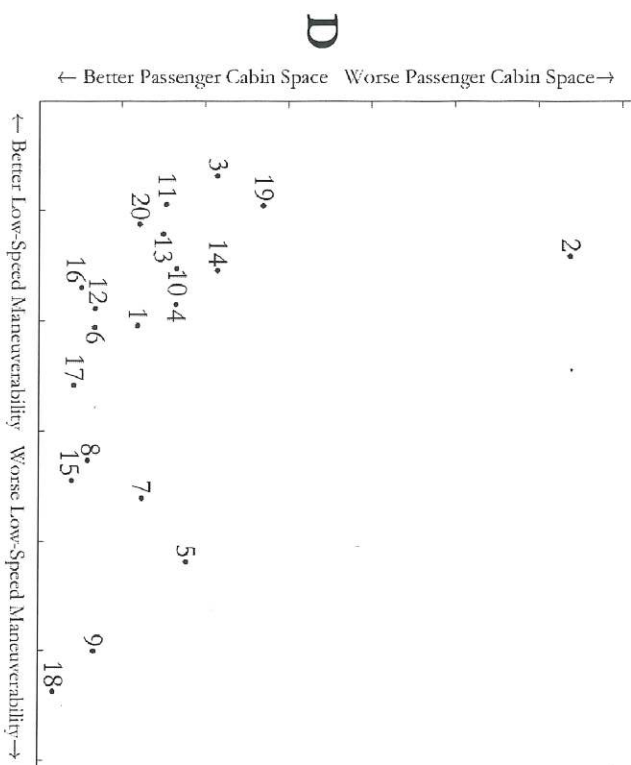
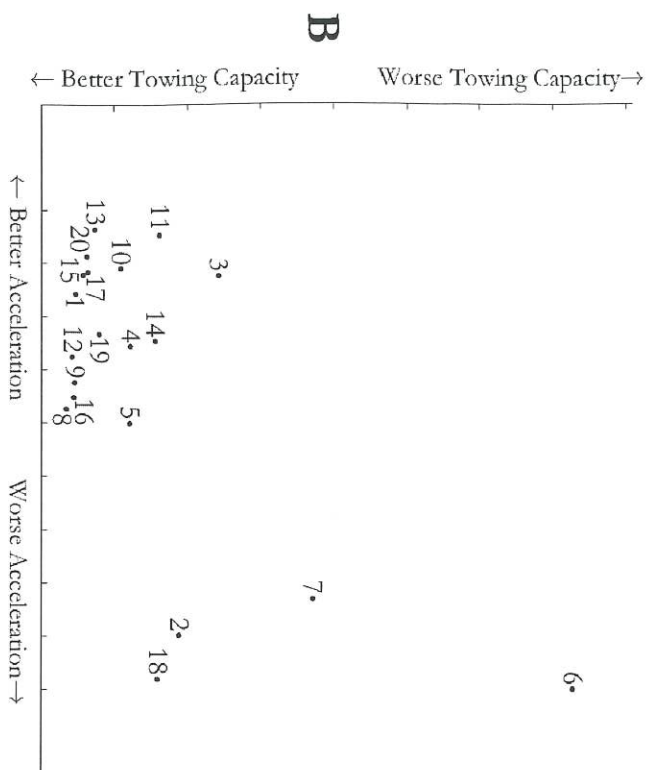
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Enter the number ID of the selected design: 17

Write your Group ID: 8050

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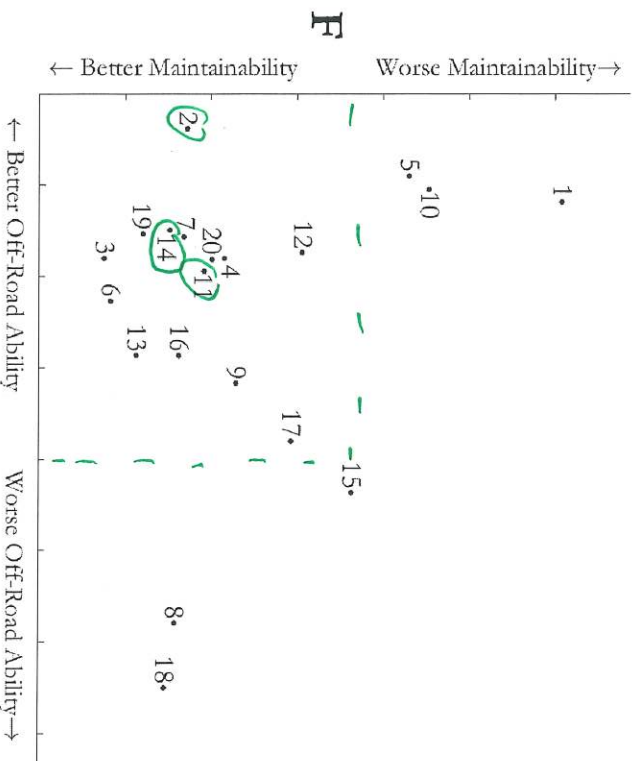
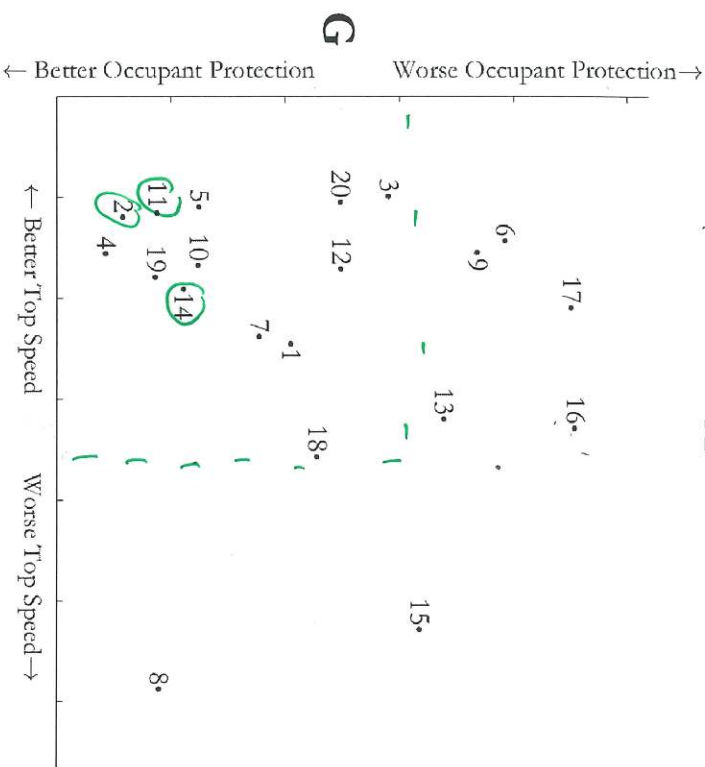
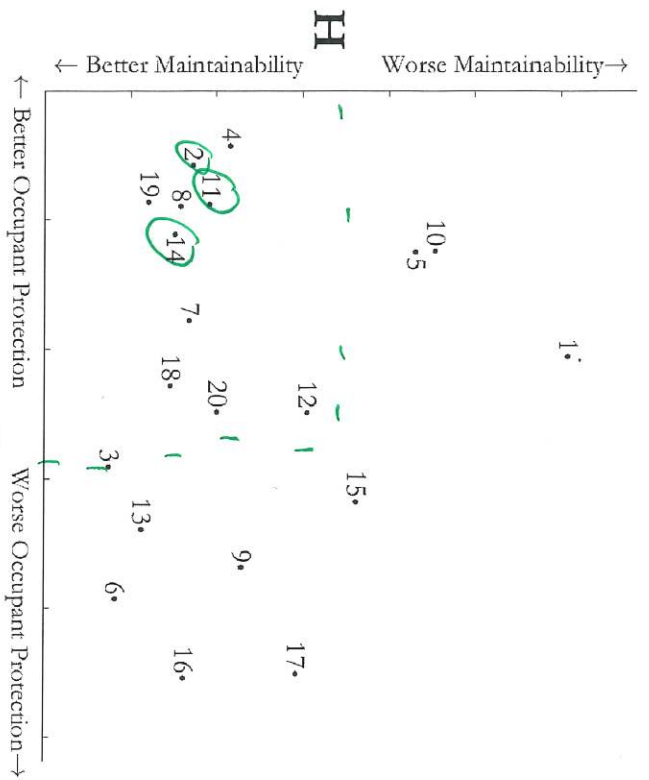
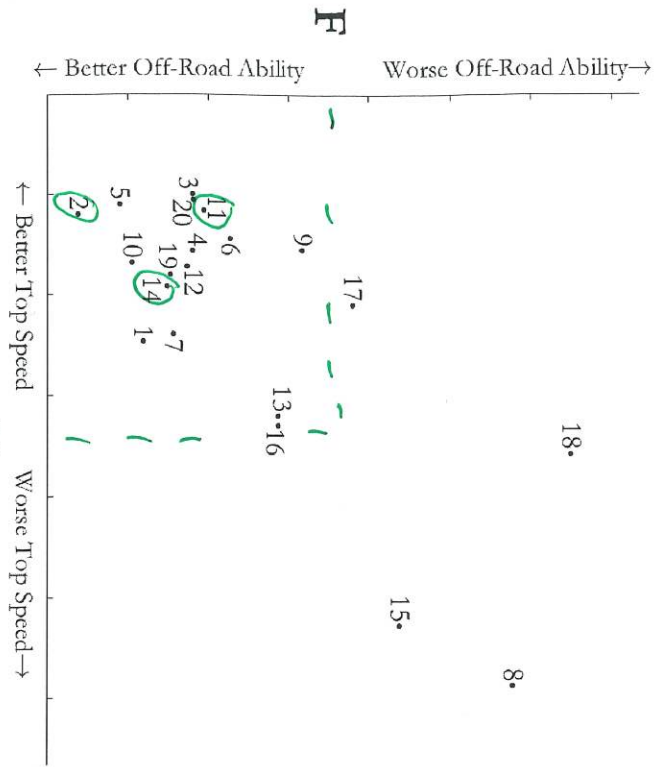
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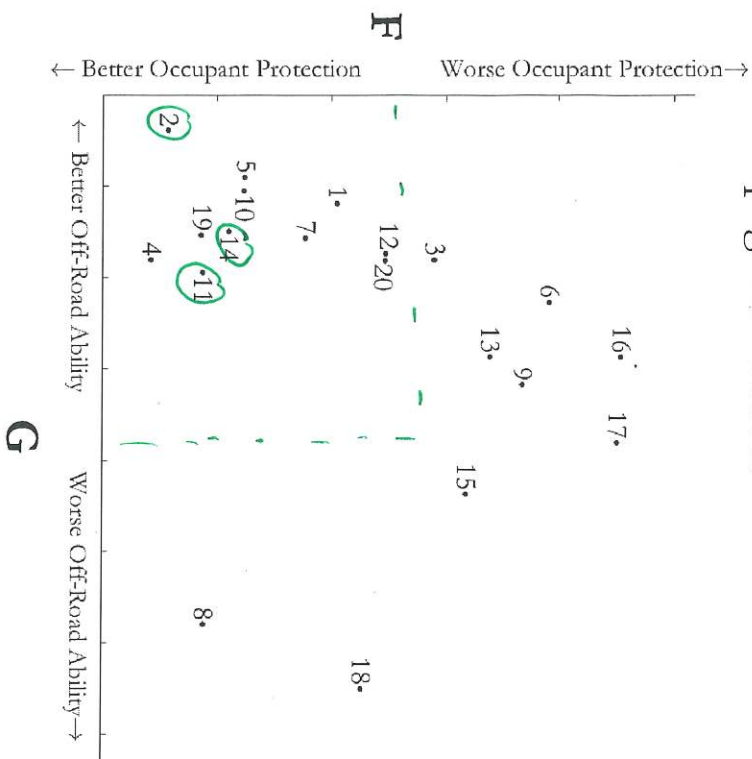
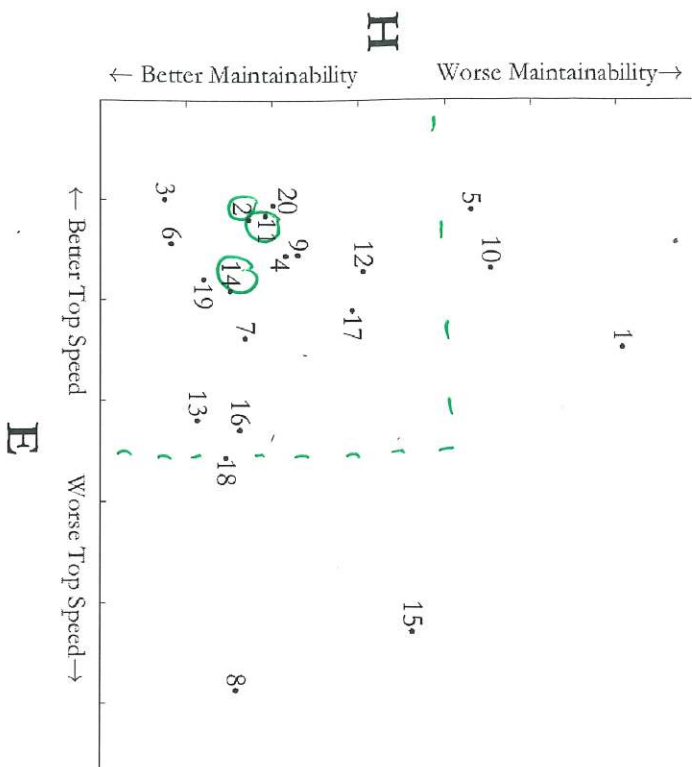
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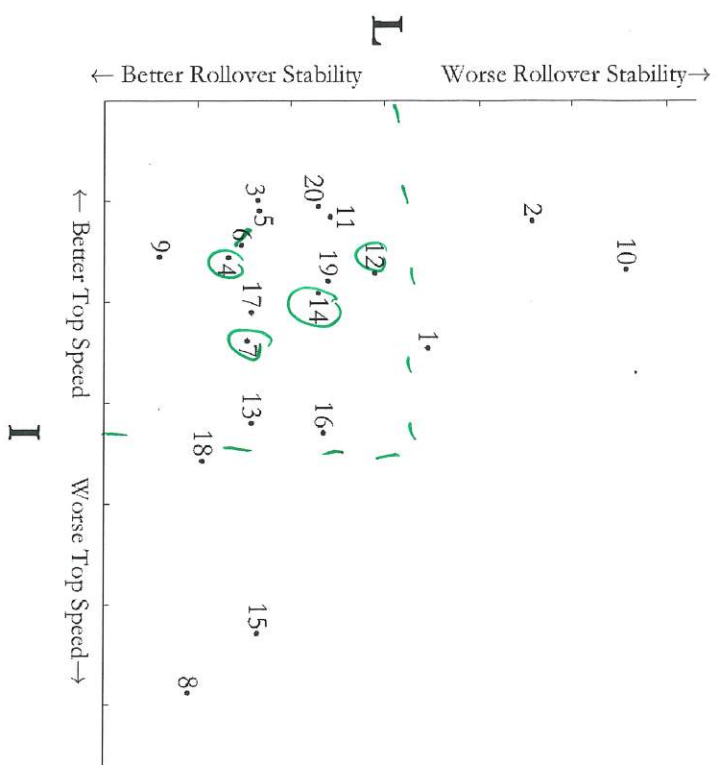
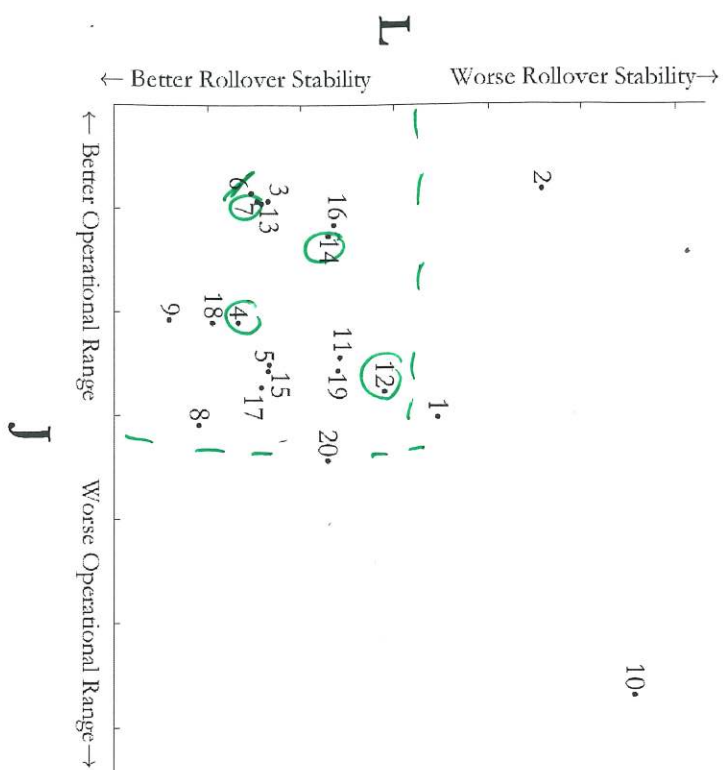
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G

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